

Robust individual alignment of color qualia structures: toward a structure-based taxonomy of divergent color experiences

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Abstract

Whether qualitative aspects of consciousness, or qualia in short, are equivalent across individuals is a foundational scientific question. Testing this is challenging because one cannot assume a shared mapping between stimuli and private experience (my “red” may be your “green”) [1–3]. Previously, we proposed a structural characterization of qualia [4, 5] and the quantitative assessment of structural correspondences through an unsupervised alignment method [4, 6], which does not presuppose such correspondence. Using this approach, our previous work focused on identifying optimal mappings between relational structures of color qualia at the group level [4]. Given known perceptual diversities [7], however, it remained unknown whether any two individuals’ structures could be empirically aligned. Here, we resolve this by collecting 4,371 pairwise similarity ratings for 93 colors from 11 individuals, enabling direct individual-to-individual alignment. We reveal

two fundamental, coexisting features. First, we identified two clusters of individuals showing robust within-cluster alignment, corresponding to color-neurotypicals and atypicals. Second, we uncovered a continuous spectrum of diversity: some participants who showed normal color discrimination ability in terms of the Total Error Score (TES) on Farnsworth-Munsell 100 hue test nevertheless failed to align with either cluster, revealing idiosyncratic structures that defy simple categorization. Together, these findings suggest a novel structure-based taxonomy of divergent color qualia that complements conventional performance-based classification. Our method is generalizable to other sensory modalities, and opens a path to the scientific investigation of both shared and idiosyncratic qualitative aspects of consciousness.

Keywords: qualia, qualia structure, consciousness, color vision, unsupervised alignment

Main

From childhood on, many of us wonder about a fundamental question: Are our subjective experiences, such as of color, the same as those of others? For instance, is my “red” the same as your “red”? Traditionally, the qualitative aspects of consciousness, or qualia for short [8, 9], have been considered unaddressable within the scientific framework due to their inherent privateness and ineffability [10, 11]. Even if two people agree that a strawberry is “red,” there is no guarantee that their internal experiences are identical [1–3].

Because it is difficult to characterize the quality of an experience in isolation, one promising approach to deal with qualia quantitatively is to focus on relational structures among qualia [4, 5, 12–27] (Fig. 1a, b). Rather than referring to absolute qualities, it is feasible to report the relationships between them, such as similarities: e.g., “red is similar to orange” or “red is dissimilar to green” (Fig. 1a). By aggregating a comprehensive set of such relational reports, we can objectively characterize and quantify the relational properties of the underlying subjective structures (Fig. 1b). We call such relational structures “qualia structures” [4]. The question of whether qualia structures are intersubjectively shared or divergent is now amenable to scientific investigation.

Crucially, comparing qualia structures requires an unsupervised approach that avoids assuming a priori correspondences between individuals’ experiences [4, 6]. Previous attempts have primarily relied on supervised methods [28–32] that assume an externally fixed correspondence between the two structures, as is typical in Representational Similarity Analysis (RSA) [33, 34] (Fig. 1c). For example, these methods presuppose that my “red” qualia maps to your “red” qualia and then quantify the degree of structural agreement under this presumed mapping. However, this “supervised” alignment is unsuitable for our purpose because there is no guarantee that the same stimulus will necessarily evoke the same corresponding color qualia in different individuals. Instead, our proposed “unsupervised” alignment method discovers the optimal mapping between two qualia structures based on their internal geometries

alone (Fig. 1d,e,f,g). Here, the optimal mapping is the maximally structure-preserving mapping [35], namely the mapping that preserves the distance relationship between color qualia across individuals to the maximum degree. By examining such mappings, we can determine whether or not it is likely that your “red” is my “red”.

Although we previously attempted an inter-personal comparison using the theoretical framework described above [4], that study left a critical question unanswered: do color qualia structures actually align across specific individuals? While we demonstrated that similarity structures align within typical and atypical groups, a group-level analysis such as this cannot determine whether structures are shared or divergent between any two specific people. Moreover, aggregating data completely masks potential diversity within each population. Statistical averaging tends to converge toward a common mean, which may obscure crucial individual differences in qualia structures. This is particularly problematic given the documented variability in color vision. Considerable diversity has been reported among color-atypical individuals at both the neurophysiological level (e.g., variability in cone spectral sensitivities [36]) and psychological level (e.g., perceived color similarities [30]). Furthermore, even among color-neurotypical observers, substantial variation in perceived unique hues [37–39] suggests that qualia structures may not perfectly align across individuals. Consequently, definitive testing for intersubjective sharing or divergence requires a direct, individual-to-individual comparison.

Here, we performed the first direct, one-versus-one individual alignment between color similarity structures, encompassing participants with diverse forms of color vision. To achieve this, we collected a complete set of 4,371 pairwise similarity judgments (Fig. 2a) for 93 colors (Fig. 2b) from 11 participants, for about 10 hours spread over 5 days per participant (Fig. 2c). Participants were assessed for their (a)typicality of color vision based on both self-report and the Farnsworth–Munsell 100 Hue Test [40] (Fig. 2d). This rich dataset is significant because it enables the first direct, individual-to-individual comparison of color similarity structures across a broad and densely sampled color space, allowing for a rigorous evaluation of how color experience varies among people. Strikingly, despite the known variability, we identified two robustly aligned groups of individuals. We refer to these post hoc as the “Typical” and “Atypical” clusters, as they closely corresponded to participants’ self-reports and their performances on the Farnsworth–Munsell test. Coexisting with this robustness, however, we observed nuanced variations in color qualia structures among certain participants, which revealed perceptual diversity not fully captured by the conventional hue test. These findings provide an empirical foundation for a structure-based taxonomy of color vision grounded directly in the phenomenology of color, while also capturing fine-grained experiential diversity that defies simple categorization. They thereby offer a novel window into the private nature of the quality of consciousness.

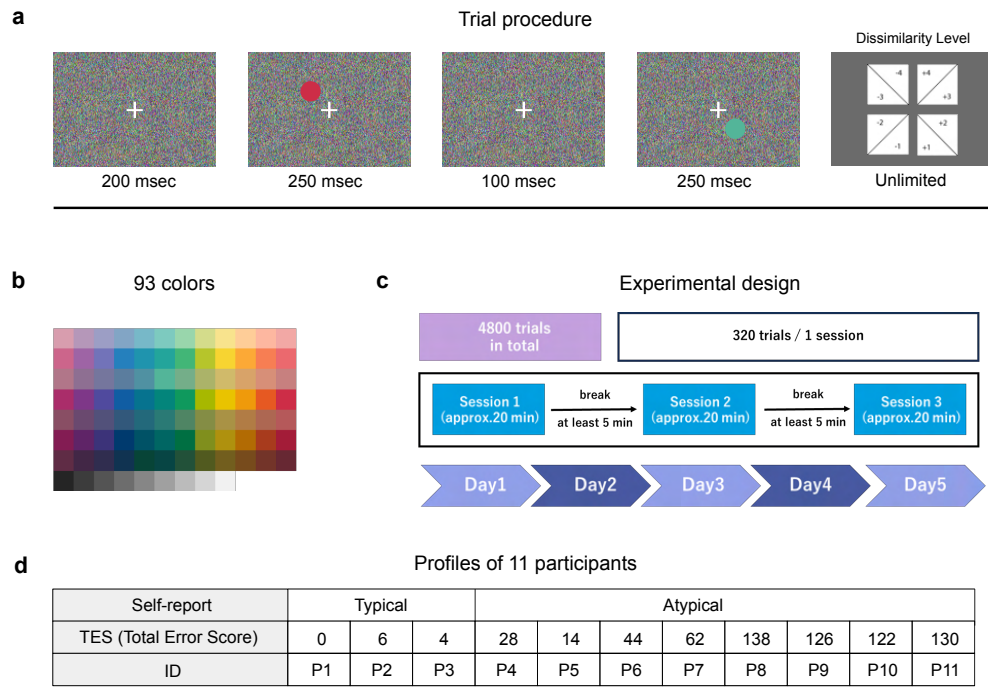


Fig. 2 Experimental procedure and participants demography **a**, Schematic of the main trial procedure (for details, see Methods). **b**, 93 colors used in our study. **c**, Experimental design of the similarity rating task. Each participant went through a complete set of similarity rating tasks for about 10 hours spread over 5 days (for details, see Methods). **d**, Profiles of 11 participants. Participants were assessed for their (a)typicality of color vision based on self-report and the FM 100 hue test. A higher TES (Total error score) indicates lower color discrimination performance on the FM100. For detailed error patterns, see Supplement Fig. S1

given known perceptual diversities. Second, we investigate whether there exists a group of individuals where structures align consistently among members. Such a finding would allow us to classify individuals based on the internal geometry of their experience rather than their performance on standard discrimination tests. If successful, these alignments would suggest that the relational geometry among qualia is shared within specific populations, providing an empirical approach to the long-standing philosophical puzzle, “Is my red your red?” [4]

To empirically identify the inter-subjective correspondence of color qualia structures, we searched for the optimal mapping between two individuals’ color similarity structures using Gromov-Wasserstein optimal transport (GWOT) [35]. This method identifies the mapping that best corresponds to the experiences of two individuals by minimizing the Gromov-Wasserstein Distance (GWD), which quantifies the distortion of distance relationships between two structures. The result is the optimal transportation plan Γ (Fig. 1f), in which each element Γ_{ik} represents the probability that the i -th color experience in one person corresponds to the k -th color experience in the other. The optimal transportation plan can be seen as a relaxed mapping that allows

for coarse many-to-many correspondence (as shown in Fig. 1g), and effectively create a translation dictionary between the qualia of two different people. This mapping is deemed “optimal” because it represents the most compatible alignment of two similarity structures among all possible graded mappings. Once this optimal mapping is established, we can directly test the equivalence of experiences by checking, for example, whether your “red” maps to my “red” (Fig. 1f, g). We quantify the degree of this correspondence using the top- k matching rate, which measures the probability that colors are consistently mapped across individuals (see Methods).

Complete collection of similarity judgments for 93 colors from 11 participants

To perform one-versus-one unsupervised alignment across individuals, we collected a complete dataset of 4,371 pairwise similarity judgments for 93 colors from 11 participants (Fig. 2a; see Methods). The colors were selected from the Practical Color Coordinate System (PCCS) [41], following the stimulus set used in [42]. The final set comprised 12 hues sampled across 7 tones, along with 9 achromatic colors ranging from white to black (Fig. 2b). Given that completing 4,371 judgments requires substantial time and mental effort, we distributed the experiment over five days, with approximately two hours of data collection per day (Fig. 2c).

11 participants (P1–P11) were assessed for the (a)typicality of their color vision based on both self-report and the Farnsworth–Munsell 100 Hue Test (FM100) (Fig. 2d). In our recruitment, we explicitly stated that we sought participants with both typical and atypical color vision, with a particular emphasis on recruiting individuals with atypical color vision. During the screening process, prospective participants were asked to self-report whether they were aware of having typical or atypical color vision using a two-alternative forced-choice question. Based on these self-reports, we recruited 3 participants with self-reported typical color vision (P1–P3) and 8 participants with self-reported atypical color vision (P4–P11). In addition to self-reports, all participants completed a standard color vision assessment, the Farnsworth–Munsell 100 Hue Test (FM100) [40]. Performance on the FM100 is summarized by the Total Error Score (TES), with higher scores indicating lower color discrimination (Fig. 2d). As a reference point, it has been reported that, among color-neurotypical individuals, approximately 16% score $\text{TES} \leq 16$, while another 16% score $\text{TES} > 100$ [43]. Note that the TES on FM 100 alone does not allow for a definitive distinction between typical and atypical color vision. Detailed error patterns on FM100 for all participants are shown in Supplementary Fig. S1.

Unsupervised alignment reveals robust clusters of shared color qualia structure, corresponding to Typical and Atypical clusters

By applying the unsupervised alignment method to all pairwise combinations of participants, we identified two data-driven clusters of individuals, each of which showed robust within-group alignment. While we visualize representative alignments below,

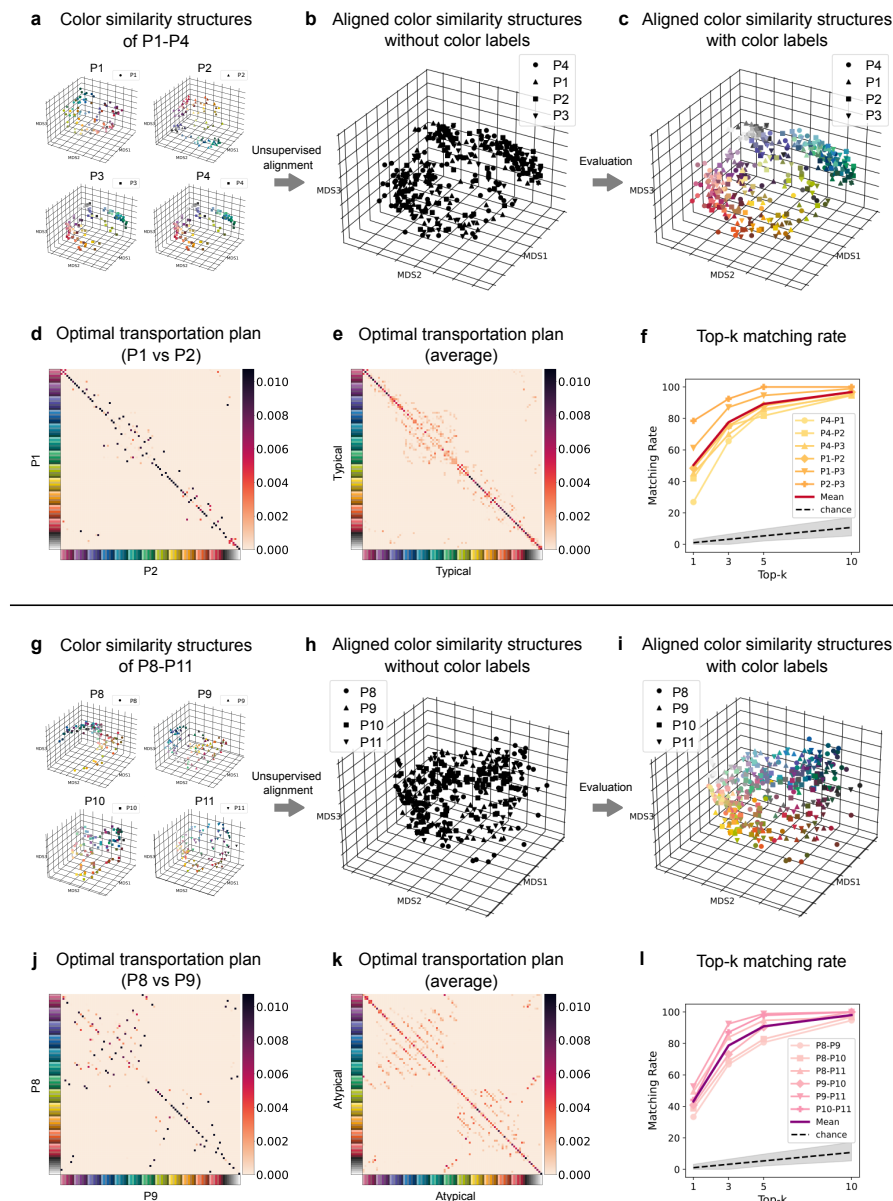


Fig. 3 Individuals form two robustly aligned groups corresponding to color-neurotypicals and atypicals. **a**, Color similarity structures of four participants, P1, P2, P3, and P4. Structures were first estimated by applying 5D multidimensional scaling (MDS) to individual similarity ratings. For visualization, we reapplied 3D MDS to the resulting 93×93 distance matrices (see Methods). Axes represent the MDS dimensions. Each point represents a color experience, and the Euclidean distance between points reflects the perceived dissimilarity between the two colors. **b**, Aligned color similarity structures of P1–P4. Color similarity structures of P1, P2 and P4 are overlaid onto P3’s color similarity space, using the optimal transportation plan Γ (see Methods). Note that the alignment procedure is performed in a purely unsupervised manner without reliance on color labels. **c**, Aligned color similarity structures of P1–P4, colored with the external color labels. **d**, Optimal transportation plans Γ between P1 and P2. Rows and columns correspond to the 93 colors of two individuals, respectively. The (i, j) element shows the probability that color i in one individual is mapped to color j in the other. **e**, The average optimal transportation plan over all pairs of P1–P4, calculated as the average of all the optimal transportation plans and their transported matrices. **f**, Top- k matching rate across over all pairs of P1–P4. For each pair, the values of top- k matching rate is plotted against the value of k (1, 3, 5, and 10). The highlighted line represents the averaged matching rate across all pairs. The dashed line represents the chance level, and gray-shaded region represents the 95% interval of the chance level (see Methods). **g–l**, Same format as a–f, but for the four participants P8, P9, P10, and P11.

a comprehensive analysis of all pairwise combinations is provided in Supplementary Fig. S2. The first cluster consisted of P1, P2, P3, and P4. Only post hoc did we observe that all four participants in this cluster (P1–P4) exhibited relatively low FM100 Total Error Scores ($TES \leq 28$). P1–P3 were also typical on self-report, with P4 being an exception in terms of self-report. This pattern motivated the descriptive label “Typical cluster.” A second, equally robust cluster comprised P8, P9, P10, and P11. All four participants exhibited relatively high FM100 Total Error Scores ($TES > 100$) and self-reported atypical color vision, which motivates the descriptive label “Atypical cluster,” again post hoc.

We first visually demonstrate the results of unsupervised alignment within the aligned Typical cluster. Fig. 3a shows the estimated color similarity structures of P1–P4, where the distance between two colored points reflects the subjective dissimilarity of these two color experiences for each participant. After obtaining the color similarity structures of each participant, we performed GWOT to find the optimal transportation plan Γ between each pair. Using this plan, we rotated and flipped the color similarity structures of P1, P2 and P4 and overlaid them onto that of P3 to align them as much as possible (see Methods and Supplementary Movie 1) (Fig. 3b). Crucially, this entire alignment procedure was performed in a purely unsupervised manner without relying on color labels. We emphasize this fact by rendering all points in Fig. 3b as black. Subsequently, to evaluate the correspondence (Fig. 3c), we colored each point according to its physical stimulus label (Fig. 2b). This visualization intuitively captures the robust alignment (i.e., top-1 matching rate of 48%) among these four participants given that points of similar color, which correspond to each participant’s qualia evoked by the color stimuli, cluster tightly together in this common representation space.

Examination of the optimal transportation plans Γ (Fig. 3d) reveals the fine-grained nature of this structural correspondence, and confirms a high degree of consistency in color mappings across individuals (P1–P4). The example optimal transportation plan (Fig. 3d: P1 vs P2) exhibits high values along many diagonal elements, indicating that many colors are mapped to the same colors across individuals. This trend is confirmed by the average optimal transportation plans Γ across all pairs of P1–P4 (Fig. 3e), where most diagonal elements clearly show high values. This tendency is reflected in consistently high matching rates across all pairwise combinations of P1–P4 (Fig. 3f), all of which fall outside the 95% percentile intervals of the chance-level estimated by generating random transportation plans (see Methods). For example, in the alignment between P1 and P2, the top-1, -3, -5, and -10 matching rates are 48%, 75%, 88%, and 97%, respectively, compared with chance-level expectations of 1.1%, 3.2%, 5.4%, and 10.7%. On the other hand, the optimal transportation plans (Fig. 3d, e) exhibit off-diagonal values around bluish and greenish hues, revealing systematic confusions in these color ranges. These swapped mappings likely arise from the subjective resemblance among these colors (Fig. 3a). Taken together, these color similarity structures robustly align within this specific subset of participants corresponding to the Typical cluster, while allowing slight divergence around bluish and greenish hues.

Similarly, we observed robust unsupervised alignment within the Atypical clusters (P8–P11). This convergence was far less predictable than that for the Typical cluster,

given that “atypical” color vision encompasses a diverse range of conflicting physiological mechanisms. Nevertheless, we found that the structures of P8–P11 (Fig. 3g) could be overlaid into a common space with high precision (Fig. 3h, i). Overall, the same or similar colors from different individuals were located close to each other, although some confusion between dissimilar colors was also observed (Fig. 3h). The optimal transportation plan Γ (Fig. 3j, k) further reveals detailed aspects of this structural correspondence. The matching rate calculated from the optimal transportation plans falls outside the 95% percentile interval of the chance-level across all pairs (Fig. 3l), indicating the shared geometry in their color space. For example, in the alignment between P8 and P9 (Fig. 3j), top-1, -3, -5, and -10 matching rates are 33%, 67%, 80%, and 95%, respectively. Notably, however, swapped mappings were observed in certain color ranges, for example among purplish, greenish, and achromatic hues (Fig. 3j). These mismatch patterns were consistently observed across all pairs of participants (Fig. 3k). These colors are subjectively similar for these participants (Fig. 3g), and this similarity is reflected in the optimal transportation plan as such swapped mappings (see the case of the typical cluster Fig. 3e for comparison). Taken together, these results indicate that color similarity structures can robustly align this specific subset of participants corresponding to the Atypical cluster. As we elaborate in the Discussion, this likely reflects a shared physiological subtype (e.g., deutan) within this specific sample rather than any universal property of atypical color vision.

Finally, we confirmed that these emerging clusters correspond to structurally distinct groups by demonstrating that they do not align robustly with one another. In contrast to the robust within-group consistency, the color similarity structures fail to align between the Typical cluster (P1–P4) and Atypical cluster (P8–P11) (Fig. 4a). For example, the optimal transportation plan between P2 and P9 (Fig. 4b) contains many off-diagonal elements, and its matching rate (top-1: 1.1%, top-3: 1.1%, top-5: 3.2%, top-10: 6.5%) is essentially at chance (top-1: 1.1%, top-3: 3.2%, top-5: 5.4%, top-10: 10.7%), reflecting substantial cross-mappings across a wide range of colors. Notably, however, some individual pairs (e.g., P1 vs. P8) exhibit moderate alignment (Fig. 4c, d). Although their matching rates (top-1: 7%, top-3: 11%, top-5: 17%, top-10: 36%) remain below those observed within the Typical or Atypical clusters, they nonetheless exceed the 95% percentile interval of the chance-level, indicating coarse alignment across the two groups. These coexisting features of strong divergence and moderate alignment are systematically captured in the matching rates across all Typical–Atypical pairs (Fig. 4e). These results indicate that while the two clusters possess distinct color similarity structures, they retain broad geometric commonalities that allow for coarse alignment in specific participant pairs.

Co-existence of robust clusters and a continuous spectrum of diversity in color qualia structures

Beyond the robust Typical and Atypical clusters, we identified a subset of participants (P5, P6, and P7) whose color similarity structures did not robustly align with either cluster. For example, P6 did not align with a Typical participant (P1: Fig. 4f, g) or an Atypical participant (P8: Fig. 4h, i), with the optimal transportation plans

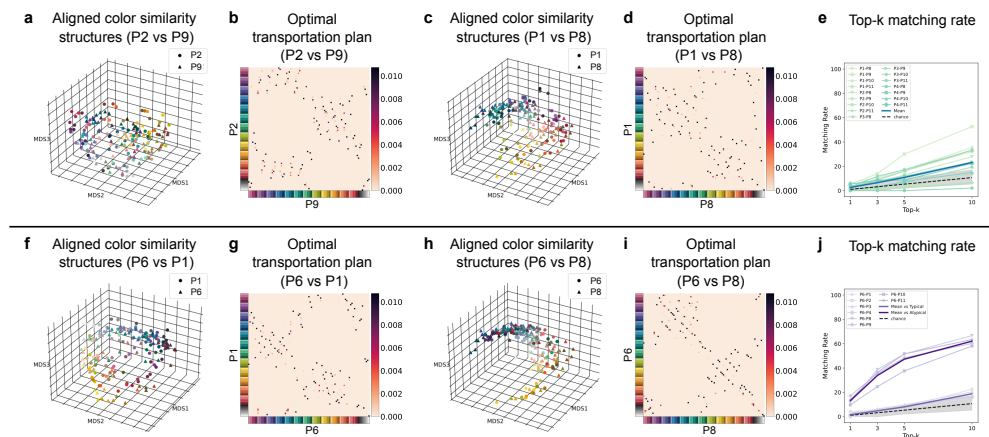


Fig. 4 Certain individual pairs fail to align strictly under unsupervised alignment. a-b, c-d, f-g, h-i, Aligned color similarity structures of P2 and P9, P1 and P8, P6 and P1, and P6t and P8 in the same format as Fig. 3c and d, respectively. e, j Top- k matching rate across over all pairs between e) Typical (P1–P4) and Atypical (P8–P11) clusters and j) across over all pairs including P6. In panel j, mean matching rates for the Typical cluster and Atypical cluster are plotted separately .

of each pair showing dispersed probability mass and low matching rates. This pattern is generalized across other pairwise comparisons involving P6 (Fig. 4j). Although P6 showed relatively stronger alignment with the Atypical cluster, these matching rates remained below those observed within either cluster, thereby indicating that P6 possesses a unique color similarity structure distinct from both clusters. Similarly, P5 and P7 did not show robust alignment with either the Typical or Atypical clusters. Notably, although FM100 Total Error Scores of participants P5–P7 fell within the range commonly observed among color-neurotypical observers [43], they nonetheless self-reported having atypical color vision. Consistent with these self-reports, the unsupervised alignment indicated idiosyncratic color similarity structures in these participants.

Combining the results from the previous sections, the full 11×11 individual alignment reveals a dual structure characterizing color qualia: the coexistence of robust clusters and a continuous spectrum. To systematically illustrate this global landscape, we examined the structural relationships between individuals through two analytical lenses: the direct quantification of pairwise matching rates (Fig. 5a) and a meta-MDS embedding that maps the global structural distances between participants (Fig. 5b). In this meta-MDS, each individual is placed in a two-dimensional space such that the distances between points reflect how distant these individuals are, quantified as the difference in the entire histogram patterns separately for $k=1, 3, 5$, and 10 (see Methods). As expected from the preceding analyses, the Typical (P1–P4) and Atypical (P8–P11) clusters show high peaks within their respective groups (yellow-red and pink-purple bars, respectively, in Fig. 5a). In contrast, P5–P7 exhibit transitional profiles. Specifically, P5 aligns moderately with both clusters with slightly better alignment

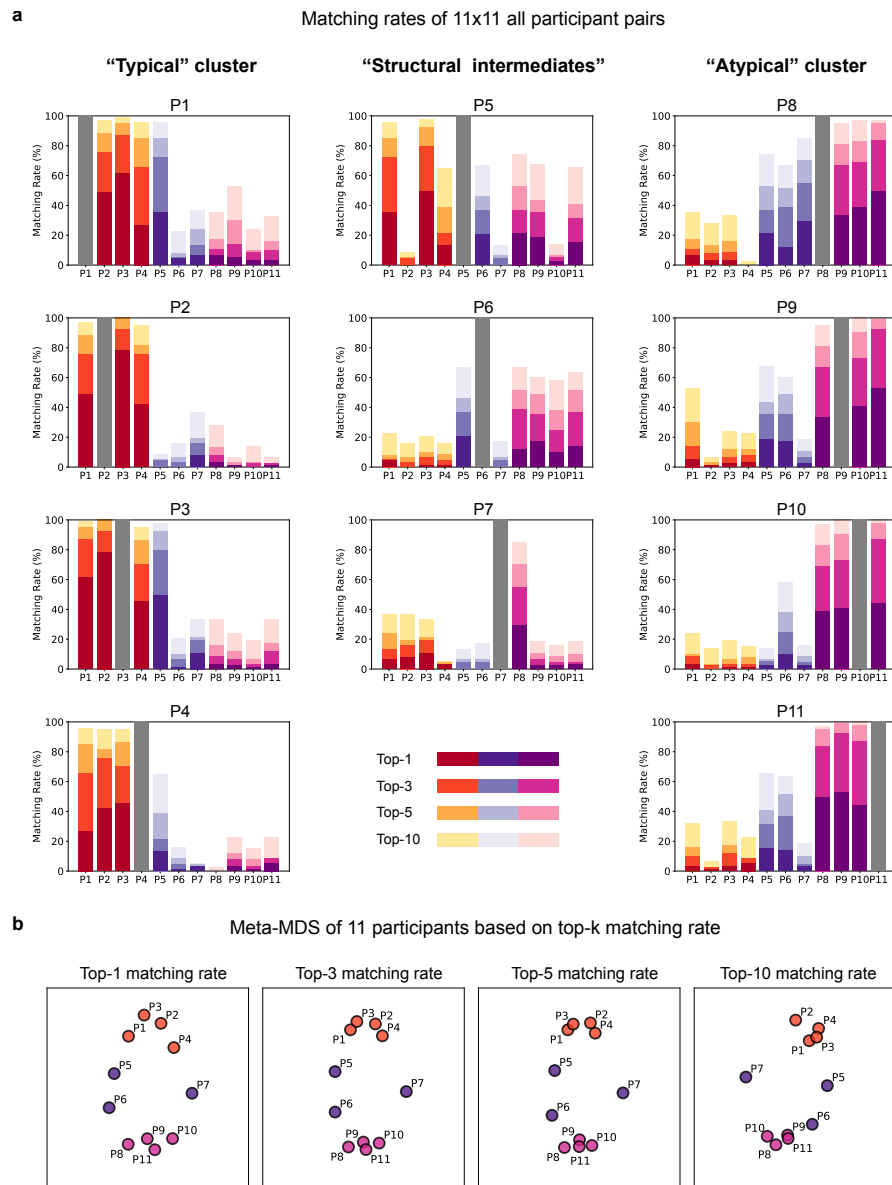


Fig. 5 Unsupervised alignment reveals a continuous spectrum of diversity in color similarity structures. **a**, Matching rates across all 11×11 participant pairs. Each panel shows the matching rate of one individual with every other participant. Notably, participants P5–P7, who passed the FM test but identified themselves as having atypical color vision, show transitional matching-rate profiles that bridge the two robust clusters, namely the Typical of P1–P4 and Atypicals of P8–P11. **b**, Meta-MDS of 11 participants based on top- k matching rate. Each plot is made by performing 2D MDS using a distance matrix calculated from the top- k matching rate separately for $k = 1, 3, 5$, and 10, where distance between participants is computed as the mismatch between histograms (see Methods).

with typical; P6 aligns more strongly with the Atypical cluster while remaining distinguishable from it due to lower matching rates; and P7 robustly aligns with neither cluster, except for its alignment with P8. Consistent with these profiles, the meta-MDS places P5–P7 along a continuum bridging the Typical and Atypical clusters rather than forming a distinct group (Fig. 5b), thereby visually confirming the coexistence of categorical clusters and continuous intermediates, which we refer to as “structural intermediates”.

Collectively, these findings point toward a structure-based taxonomy of color vision grounded directly in the phenomenology of color experience. Our analysis first identified that color similarity structures form two robustly aligned clusters, corresponding to the “Typical” (P1–P4) and “Atypical” (P8–P11) clusters in our descriptive label. Nevertheless, against this baseline of within-cluster convergence, we revealed that a subset of the population, or the “structural intermediates” (P5–P7), resists simple categorization, but rather distributes continuously between these two poles (Fig. 5b). By capturing this fine-grained structural variation, our unsupervised alignment approach provides a higher-resolution framework for categorizing subjective experiences, revealing individual differences or commonalities that are not fully captured by standard performance-based metrics.

Discussion

In this study, we demonstrate that color qualia structures can be robustly aligned between individuals in a purely unsupervised manner, and reveal both shared structure and systematic divergence across people. From 4,371 pairwise similarity judgments for 93 colors obtained from 11 individuals with diverse forms of color vision, we revealed dual, coexisting features in color qualia across individuals. Strikingly, we identified two robustly aligned groups of individuals corresponding to “Typical” and “Atypical” clusters, together with clear structural divergence between them. These clusters emerged entirely in a data-driven manner without reference to hue-test performance or self-reported phenomenology, yet post hoc they closely reflected both. At the same time, we also identified certain individuals who did not align with either cluster (“structural intermediates”), and thereby formed a continuous spectrum between the Typical and Atypical clusters. Notably, all of these individuals self-reported atypical color vision despite showing normal color discrimination ability in terms of TES on FM100 test. Together, these results point toward a structure-based taxonomy of color experience, offering a novel categorization that complements standard performance-based metrics.

The robust alignment within the “Typical” cluster is not an obvious observation, given the substantial individual differences documented among color-neurotypicals at both neurophysiological and perceptual levels. At the neurophysiological level, color-neurotypicals exhibit considerable variability in their photoreceptors, including a shift in cone spectral sensitivities [44] and the L:M cone ratio [45, 46]. The presence of robust alignment despite these sources of variability suggests that subsequent cortical mechanisms can accommodate such physiological differences [30, 47] and instantiate qualitatively similar color experiences based on a shared structure [48–52]. In fact, P4

exhibited an FM100 error pattern suggestive of mild anomalous trichromacy, a condition in which all three cone types are present but exhibit altered spectral sensitivities (see Supplementary Fig. S1 and Supplementary Text 1). This observation highlights a potential dissociation between physiological traits and the phenomenal structures captured by our alignment. At the perceptual level, color-neurotypicals also show large and systematic individual differences in the loci of unique hues [37–39] and in color naming, in which observers cluster into distinct lexical patterns even within the same language [53]. The seeming incompatibility of our results with these findings further suggests that these perceptual traits, including judgments of uniqueness or lexical categorization, may be dissociable from the color qualia themselves.

The robust alignment within the “Atypical” cluster may reflect a contingent outcome rather than a general property of atypical color vision, and should therefore be interpreted carefully. Generally, “atypical” color vision encompasses a diverse range of physiological mechanisms, such as protan or deutan, and varying degrees of severity [7, 54–56]. Given this well-documented heterogeneity, the strong within-group convergence seen in our Atypical cluster suggests that these individuals coincidentally share a common physiological subtype. Although we cannot directly determine the precise physiological subtypes of atypical color vision in our dataset, the characteristic error patterns observed in the FM 100 hue test allow for a cautious inference (see Supplementary Fig. S1 and Supplementary Text 1). Specifically, these patterns are most consistent with deutan-type atypical color vision [43], in which M-cone function is absent or altered. This inference is also consistent with epidemiological evidence showing that deuteranopia and related deutan conditions are more prevalent than protanopia or tritanopia (involving L-cone or S-cone absence or anomalies, respectively) [54, 56]. It should also be noted that the available psychophysical data do not permit a clear distinction between dichromacy (the complete absence of one cone class) and anomalous trichromacy (the presence of all three cone classes with altered spectral sensitivity) [57]. It is therefore plausible that, by chance, the individuals in the Atypical cluster (P8–P11) share relatively similar deutan subtypes or degrees of severity, leading to the observed within-cluster convergence.

Despite overall structural divergence between the Typical and Atypical clusters, the presence of coarse alignment in some cross-cluster pairs (Fig. 4c-e) suggests that aspects of color experiences may still be shared to some extent across these groups. The dominant view based on color models characterizes color-atypical observers as experiencing a restricted subset of neurotypical color qualia. However, a parallel line of work has long suggested that their color experiences can nonetheless be rich and structured in ways comparable to those of neurotypicals [28, 58], consistent with our observations. This view is further supported by a historical fact: in the long history of humanity, “color blindness” was not scientifically recognized until John Dalton’s seminal report in the late 18th century [59]. This delay implies that atypical color vision may have long gone unnoticed because atypical and neurotypical observers communicated without difficulty using a shared color vocabulary grounded in largely overlapping phenomenology. An alternative, less likely but still intriguing possibility is that the shared linguistic system itself shapes color experience in a way that renders their qualia similar [60].

The continuum bridging the Typical and Atypical clusters, as revealed by the presence of “Structural intermediates” (P5–P7), likely reflects the known physiological variability in color vision [54–56]. For P6 and P7, their color similarity structures were more similar to those of the Atypical cluster than to the Typical cluster, while still failing to robustly align with either cluster. This observation is consistent with their FM100 error patterns, which suggest protan-type atypical color vision (see Supplementary Fig. S1 and Supplementary Text 1), in contrast to the deutan-type patterns that were most consistent with the Atypical cluster. In addition, their Total Error Scores tended to be lower than those observed in the Atypical cluster, which aligns with their intermediate position between the Typical and Atypical clusters. Taken together, these observations suggest that P6 and P7 exhibit moderate, protan-like atypical color vision that is distinct from the pattern observed in the Atypical cluster. One intriguing possibility is that these participants correspond to anomalous trichromacy, which has been shown to span a continuous spectrum of color vision phenotypes [30, 36], although the FM100 does not allow a reliable distinction between anomalous trichromacy and dichromacy. By contrast, P5 exhibited FM100 performance within the range commonly observed among color-neurotypical observers, despite failing to robustly align with the Typical cluster. This pattern leaves open the possibility that P5 exhibits a mild form of color vision variation, such as anomalous trichromacy, that is not detectable by the FM100 but could potentially be revealed by other, more sensitive assessments.

Our findings highlight the value of a structure-based taxonomy grounded directly in the phenomenology of color experience, beyond what physiological or performance-based diagnoses can provide. Many traditional assessments of color vision, such as the FM 100 hue test and the Ishihara test, often rely on threshold-based performance measures and thus divide observers into coarse categories that may obscure the continuous nature of color vision differences. Physiological classifications likewise cannot specify how such variations manifest in subjective experience. In contrast, our structure-based classification draws solely on the qualia structures inferred from similarity judgments, enabling a phenomenology-first taxonomy of color vision. Notably, our results highlight a dissociation between performance on the FM100 and structure-based classification. For example, P5 exhibited FM100 performance within the normal color discrimination range, yet showed a color similarity structure that was distinct from the Typical cluster. Conversely, P4 exhibited an atypical-like error pattern on the FM100 but robustly aligned with the Typical cluster. These dissociations indicate that our structure-based approach can reveal phenomenological insights that are not fully captured by standard performance-based metrics.

A limitation of our study is the relatively small number of participants ($N = 11$), which constrains our ability to fully capture the continuous spectrum of variation in the broader population. This limited sample size reflects our decision to prioritize the collection of rich datasets employing a substantially larger set of color samples than in previous studies, which used fewer than 30 colors to collect color similarity ratings from each individual [28, 30–32, 61–64]. This larger set of color samples is necessary to densely sample color space across both hue and tone dimensions, which would not be possible with such small color sets. However, it remains unknown whether additional

clusters beyond the two identified here would emerge with a larger cohort (as suggested in [13]). Given the substantial practical burden of obtaining complete sets of similarity judgments for a large number of stimuli, scaling up participant numbers will likely require more efficient sampling methods that can approximate fine-grained similarity structures across a large stimulus space with fewer trials.

Our one-versus-one alignment framework is inherently generalizable, providing a principled way to capture and compare the structured organization of various aspects of subjective experience across individuals. Beyond color vision, we have already applied this unsupervised alignment method to multiple domains, including object recognition [65, 66] and emotion [67]. More broadly, the approach is particularly well suited to domains in which individual differences in phenomenal experience are central and both shared and idiosyncratic organization must be captured. These include the relationship between phenomenology and development [32], interactions with language systems [68, 69], and forms of neurodivergence such as autism [70]. Ultimately, our approach enables subjectivity itself to be treated within a scientific framework, and thereby offers a foundation for addressing the long-standing question of how conscious experiences can be shared and compared.

Methods

Psychophysics experiment

Participants

Eleven participants (10 males and 1 female; mean age = 20.9 ± 2.6 years) took part in the experiment. Participants were undergraduate and graduate students from the University of Tokyo. They were recruited via email, which explicitly stated that we were seeking individuals with both typical and atypical color vision. During recruitment, participants self-reported their color vision status: according to these reports, 3 participants had typical color vision (including the female) and 8 had atypical color vision. The experiment was approved by the Research Ethics Committee of the University of Tokyo. All participants provided informed consent prior to the experiment and received monetary compensation upon completion.

Color similarity rating task

Apparatus

The experimental stimuli were presented on a VIEWPixx monitor (Vpixx Technologies). The monitor had a width of 52 cm with a resolution of 1920 (W) $\times$ 1080 (H) pixels, and the viewing distance was set to 57 cm. The stimuli were created using PsychoPy-2023.2.3 [71]. The monitor was calibrated using a ColorCAL MKII Colorimeter (Cambridge Research Systems, Rochester, UK) and the luminance was gamma-corrected. All experiments were conducted in a dark room in which only the monitor provided illumination.

Stimuli

93 colors were selected as stimuli from the Practical Color Coordinate System (PCCS) based on those used in [4] (Fig. 2b). These 93 colors included 84 chromatic

colors, derived from combinations of 12 hues and 7 tones, and 9 achromatic colors. All color stimuli were presented as circles with a diameter of 1 deg. The stimuli were displayed on a background of 1920×1080 pixels, each having a randomly assigned RGB value. The mean RGB value of the background pixels was set to [128, 128, 128].

Procedure

Each participant rated the similarity of a total of 4,800 color pairs. These included all 4,371 ($=_{93} C_2 + 93$) possible combinations of 93 colors (including pairs of identical colors), along with an additional 429 randomly selected pairs from the 4,371 pairs (double-pass) to assess response consistency. Each session consisted of similarity ratings for 320 pairs, including 28 or 29 double-pass pairs. The experiment consisted of 15 sessions conducted over five days. The assignment of all color pairs to sessions was randomized for each participant.

In each trial, two color stimuli were presented sequentially on the screen. Participants were then asked to rate the similarity of the two colors (Fig. 2a). A fixation point was displayed at the center of the screen for 200 ms, followed by the sequential presentation of two color stimuli for 250 ms each, with an inter-stimulus interval (ISI) of 100 ms. The stimuli were randomly presented at one of four locations, offset by 1 deg both horizontally and vertically from the fixation point, with the positions of the first and second stimuli adjusted to ensure they did not overlap. The order of color pair presentation was randomized for each participant. A response screen appeared 100 ms after the presentation of the second stimuli. Participants rated the similarity of the two colors by clicking on one of the following values: [-4, -3, -2, -1, 1, 2, 3, 4]. The response scale was explained to participants as follows: [-4: Least similar, -3: Very dissimilar, -2: Dissimilar, -1: Slightly dissimilar, 1: Slightly similar, 2: Similar, 3: Very similar, 4: Most similar]. Participants were instructed to use the full range of the scale, including intermediate values, when making their ratings. After responding, participants were asked to click on a small square in the center of the screen so that they fixate the center of the cross in the next trial.

At the beginning of each session, participants completed 20 practice trials to remind themselves of the task procedure, recalibrate the similarity ratings, and adapt to the luminance level of the experimental room. During the practice, participants received feedback on their responses, which was limited to which rating they used and what the particular rating means. The results of the practice trials were excluded from analysis. During the main trials, a pause screen was displayed every 40 trials, allowing participants to take short breaks as needed. Each session took approximately 20 minutes to complete.

To evaluate the internal consistency of participants' responses, we computed a double-pass correlation for each participant by pooling all stimulus pairs that were presented twice across sessions. The double-pass correlation was defined as the Pearson correlation coefficient between the two sets of ratings given to the same stimulus pairs. This measure reflects the degree to which a participant consistently judged the similarity of identical color pairs across repeated presentations. Across participants, the number of double-pass pairs was $N = 429$, yielding a degree of freedom $DOF = 427$ for each correlation estimate. The results showed high consistency across all participants, with correlation coefficients ranging from $r = 0.65$ to $r = 0.88$ (mean $r = 0.79$).

All correlations were highly significant (all $p < 10^{-50}$), indicating that the similarity judgments were reliable and stable within individuals. Session-by-session double-pass correlations were broadly consistent with the corresponding across-session correlations, although some within-participant variability was observed (Supplementary Fig. S3). One session from participant P1 showed a near-zero correlation and appeared as a clear outlier relative to other sessions across participants. Still, we decided not to exclude this session, as no session-level exclusion criterion based on double-pass correlation had been specified a priori.

Color vision assessment by FM 100 hue test

Apparatus

The Farnsworth-Munsell 100 Hue Test (X-Rite) was employed. All experiments were conducted in a dark room, and the Pantone 3 light booth (X-Rite) was used as the light source device.

Stimuli

The test utilized 85 physical colored caps, which were divided into four trays. Each tray contained 22 or 23 caps arranged in random order.

Procedure

Participants were instructed to arrange the caps in each tray in the order of their hues. The sorting task was conducted under a standard daylight light source (D65) provided by the light booth. Although the original FM 100 Hue Test specifies Illuminant C as the standard light source, the difference between Illuminant C and D65 has been shown to have a negligible effect on FM 100 Hue Test performance [72]. Participants performed two session in total, each consisting of one sorting task per tray (four trays in total). The time required to sort one tray was estimated at approximately two minutes, but no time limit was imposed for the task.

Analysis

Each colored cap was numbered sequentially from 1 to 85 based on its hue. For each cap, a score was calculated as the sum of the absolute differences in hue number between the cap and its adjacent caps. For example, if the caps were arranged in the order 10, 11, 12, 16, and 14, the score for cap 11 would be $\|10 - 11\| + \|12 - 11\| = 2$, and the score for cap 16 would be $\|12 - 16\| + \|14 - 16\| = 6$. For each cap, the adjusted score was calculated by subtracting 2 from its original score, and the total sum of these adjusted scores across all caps was calculated as the Total Error Score (TES). In addition to this summary measure, cap-wise error scores were examined to characterize the distribution of errors across hues (Supplementary Fig. S1). Based on the error patterns, the subtype of atypical color vision was inferred according to the error patterns depicted in [43] (see Supplementary Text 1).

Person-to-person unsupervised alignment between color similarity structures

Construction of color similarity structures

First, we estimated the color similarity structure of each participant using the similarity ratings obtained from the psychophysics experiment (Fig. 2a). Assuming that the

distance relationships between color experiences can be modeled within a Euclidean space, we estimated the color embeddings for each participant using Multidimensional Scaling (MDS). Specifically, for the i -th color ($i \in 1, 2, \dots, 93$), the embedding x_i in a 5-dimensional Euclidean space was computed by minimizing the Stress value defined by the following equation:

$$\text{Stress}(x_1, x_2, \dots, x_{93}) = \sum_{i \neq j=1, \dots, 93} (d_{ij} - |x_i - x_j|)^2 \quad (1)$$

Here, d_{ij} denotes the dissimilarity rating between stimuli i and j , while $|x_i - x_j|$ denotes the Euclidean distance between the embeddings x_i and x_j . The dissimilarity values d_{ij} were obtained by converting the raw similarity ratings in the range of $[-4, -3, -2, -1, +1, +2, +3, +4]$ into integer scores in the range of $[7, 6, 5, 4, 3, 2, 1, 0]$ in this order, where higher values indicate greater dissimilarity. For example, the rating of -4 (“least similar”) was converted to 7, and the rating of $+1$ (“slightly similar”) was converted to 3. For one participant (P8), a single response value (one of all possible combinations of colors) was missing, and this missing dissimilarity rating was imputed with a neutral value of 3. For color pairs included in the double-pass trials, we averaged the ratings from the first and the second presentations. This process corresponds to finding the embeddings of all colors such that the distances between them approximate the dissimilarity ratings as closely as possible.

In all subsequent analyses, we used the estimated distance structures rather than the raw similarity ratings. For each participant, 93×93 color dissimilarity matrix D was computed based on the Euclidean distances between the embedded colors, where each element D_{ij} represents the subjective dissimilarity between the i -th and j -th color experiences: $D_{ij} = |x_i - x_j|$. Although we obtained similarity ratings for all possible color pairs, we relied on the distances derived from the estimated color similarity structures for two reasons. First, we assume that the dissimilarities between color experiences are inherently continuous within each participant, whereas the collected ratings were discretized by the response format. Second, since each color pair was rated only once (except for the double-pass trials), occasional response errors (e.g., accidental button presses) could occur. However, such local noise would be effectively attenuated through the embedding procedure, which aggregates information across all pairwise ratings to find a globally consistent geometric configuration.

For visualization, we applied MDS to the distance structure of the 5-dimensional embeddings and retained the first three dimensions to obtain a lower-dimensional representation of the color similarity structure.

Gromov-Wasserstein Optimal Transport (GWOT)

To evaluate the structural correspondence between two individuals’ color similarity structures, we employed Gromov-Wasserstein Optimal Transport (GWOT). GWOT provides an unsupervised framework for identifying the maximally structure-preserving mapping between two metric spaces, in this case, two participants’ color similarity structures, denoted by color dissimilarity matrices D and D' . The correspondence is obtained by minimizing the Gromov-Wasserstein Distance (GWD),

which quantifies the discrepancy between pairwise distance relationships across the two spaces:

$$\text{GWD} = \min_{\Gamma \in \Pi(\mathbf{p}, \mathbf{q})} \sum_{i,j,k,l} (D_{ij} - D'_{kl})^2 \Gamma_{ik} \Gamma_{jl} \quad (2)$$

Here, $\Pi(\mathbf{p}, \mathbf{q})$ denotes the feasible set of transportation plans Γ , such that $\Gamma_{ik} \geq 0$, $\sum_k \Gamma_{ik} = p_i$, and $\sum_i \Gamma_{ik} = q_k$, where $\mathbf{p}$ and $\mathbf{q}$ are the marginal distributions over the two spaces. In our setting, both marginals were uniform, i.e., $p_i = q_k = 1/93$ for all i, k , imposing an equal-mass correspondence between all colors.

Because the optimization problem is non-convex, multiple local minima arise depending on initialization. To efficiently search for a good local optimum, we add an entropy-regularization term to the objective function of GWOT as follows:

$$\text{GWD}_\epsilon = \min_{\Gamma \in \Pi(\mathbf{p}, \mathbf{q})} \sum_{i,j,k,l} (D_{ij} - D'_{kl})^2 \Gamma_{ik} \Gamma_{jl} - \epsilon H(\Gamma) \quad (3)$$

where $H(\Gamma)$ is the entropy of a transportation plan Γ and ϵ is a hyperparameter that determines the strength of the entropy regularization.

To approximate the global optimum, we performed the optimization 500 times with random initializations of Γ and different ϵ values for each participant pair and picked the solution yielding the lowest GWD value without the entropy-regularization term (Eq. 2). Each element in the initial transportation plan was sampled from the uniform distribution $[0, 1]$, and normalized to satisfy the condition of the feasible set of transportation plans $\Pi(\mathbf{p}, \mathbf{q})$. We sampled 500 different values of ϵ ranging from 0.01 to 0.1. Prior to applying GWOT, the two compared dissimilarity matrices were histogram-matched to equalize their overall scale, because GWOT evaluates structural correspondence only up to isometry and is, therefore, sensitive to differences in scale between the two spaces. This optimization was implemented using the Gromov-Wasserstein Optimal Transport Toolbox (GWTune) [6].

Evaluation of the unsupervised alignment

To quantify the degree to which two individuals' color similarity structures were aligned, we computed the Matching Rate between two structures. Matching Rate represents the proportion of colors that were mapped to the same physical color under the optimal transportation plan Γ obtained by GWOT. To prepare for the computation of Matching Rate, we denote the color labels in participant 1 and 2 as c_1 and c_2 , respectively. For each color i in participant 1, denoted by c_{1i} , the matching condition is defined as:

$$\text{Match}(i) = \begin{cases} 1, & \text{if } \Gamma_{ij} = \max_{j \in \{1, 2, \dots, n\}} \Gamma_{ij} \text{ and } c_{1i} = c_{2j}, \\ 0, & \text{otherwise.} \end{cases} \quad (4)$$

This function indicates whether the i -th color in participant 1 corresponds to the same physical color in participant 2 under the optimal mapping. The Matching Rate is then given by the proportion of colors in participant 1 that match with the same physical colors in participant 2:

$$\text{Matching Rate} = \frac{1}{N} \sum_{i=1}^N \text{Match}(i) \quad (5)$$

where $N(= 93)$ is the total number of colors. A higher Matching Rate indicates stronger structural alignment between the two color similarity structures, whereas a lower value suggests greater individual divergence.

The matching rate defined above is the top-1 matching rate. More generally, we also define top- k matching rate, which evaluates more coarse alignment between two structures. For each color i in participant 1, we can define a function to determine if the probability of the i -th color corresponding to the same color in participant 2 is within the top- k probabilities:

$$\text{Top}_k(i) = \begin{cases} 1, & \text{if } \Gamma_{ij} \text{ is the top-}k \text{ for } j \in \{1, 2, \dots, n\} \text{ and } c_{1i} = c_{2j}, \\ 0, & \text{otherwise.} \end{cases} \quad (6)$$

The top- k matching rate was calculated as

$$\text{Top-}k \text{ Matching Rate} = \frac{1}{N} \sum_{i=1}^N \text{Top}_k(i) \quad (7)$$

The chance-level matching rate was estimated by randomly generating 10,000 matrices satisfying the constraints of a transportation plan. For each matrix, the matching rate was computed, yielding a null distribution of matching rates. Finally, the 2.5th–97.5th percentiles of this distribution were taken as the 95% percentile interval for the chance-level matching rate.

Visualization of the unsupervised alignment

We visualized the results of the unsupervised alignment to intuitively demonstrate how the color similarity structures from different participants are aligned in a shared space. Here, we used the color embeddings X and Y and the optimal transportation plan Γ that had been obtained in the previous analysis steps. Each embedding matrix $X, Y \in \mathbb{R}^{d \times n}$ (with $d = 5, n = 93$) represents the color similarity structure for one participant, where each column corresponds to a color and each row corresponds to one embedding dimension. Thus, the distances among columns in X and Y reflect the perceptual dissimilarity between colors estimated from individual similarity judgments. Multiplying Y by the optimal transportation plan Γ from the right (i.e., $Y\Gamma$) yields a transformed embedding in which each column of $Y\Gamma$ represents a weighted combination of the original embeddings in Y , according to the correspondence encoded in Γ . In the special case where Γ is a permutation matrix, this transformation reduces to a simple reordering of the columns of Y in accordance with the one-to-one correspondence specified by Γ .

To align the two structures X, Y in the same coordinate system, we searched for an orthogonal transformation matrix Q that rotates and reflects $Y\Gamma$ so that its overall geometric configuration best overlaps with X . This motivation leads to the following

orthogonal Procrustes problem:

$$\min_Q \|X - QY\|_F^2 \quad (8)$$

where $\|\cdot\|_F$ denotes the Frobenius norm. The optimal Q was obtained via the singular value decomposition (SVD) of $X(Y^\top Y)^{-1}Y^\top$. The aligned color embeddings were then visualized in a common space using X and QY .

Meta-MDS of 11 participants based on top- k matching rate

To visualize how the 11 individuals resemble each other or diverge in their color similarity structures, we embedded participants into a two-dimensional space using the following procedure. First, for each participant, we computed a histogram of the top- k matching rates obtained from pairwise alignments with all participants, where the self-matching rate was fixed at 100% (as shown in Fig. 5a). The dissimilarity between a pair of participants was then defined as the L_1 distance between their corresponding top- k matching-rate histograms; that is, we subtracted one histogram from the other and summed the absolute values of the resulting differences. Using the resulting participant-by-participant distance matrix, we performed two-dimensional multidimensional scaling (MDS) and plotted each participant in the resulting embedding space. This entire procedure was performed separately for $k = 1, 3, 5$, and 10.

Supplementary information.

- **Supplementary Text 1.** Classification and interpretation of FM100 hue test results.
- **Supplementary Fig. S1.** Error patterns on the FM 100 Hue Test for 11 participants.
- **Supplementary Fig. S2.** Unsupervised alignment of all 11×11 participant pairs.
- **Supplementary Fig. S3.** Double-pass correlations for 11 participants.
- **Supplementary Movie 1.** Unsupervised alignment of four participants corresponding to the Typical clusters (P1–P4). <https://www.youtube.com/watch?v=6MCVo7Ssd0I>

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test results. Y.T. and M.O. drafted the initial manuscript. All authors reviewed, edited, and approved the final manuscript.

Competing interests. The authors declare no competing interests.

Data and code availability. The datasets generated and/or analyzed during this study and the custom computer code used to generate the reported results are available from the corresponding author upon reasonable request during peer review and will be made publicly available upon publication.

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Supplementary Materials

Supplementary Text 1: Classification and interpretation of FM100 hue test results.

We inferred both the presence of atypical color vision and the subtype of atypicality based on cap-wise error patterns described in previous work on the Farnsworth–Munsell 100 Hue Test (FM100) [43]. We emphasize at the outset that these inferences are not intended as definitive categorizations of underlying physiological subtypes, but rather as descriptive classifications based on characteristic error patterns observed in the FM100. To infer the color vision subtype, we examined the distribution of cap-wise errors in the circular plot. Although errors were present across the full circle, we focused primarily on the right half of the plot (cap numbers 44–85), where the protan and deutan confusion axes are most clearly differentiated, whereas these axes are substantially closer in the left half and therefore less informative for subtype discrimination. Specifically, when the dominant error peak overlapped with the protan confusion axis but not with the deutan confusion axis, we inferred protan-type atypical color vision. When the dominant error peak overlapped with the deutan confusion axis, we inferred deutan-type atypical color vision, regardless of any concurrent overlap with the protan axis. Participants who did not show a clear overlap with either confusion axis were inferred to be color-neurotypical.

Applying these criteria, we inferred participants P8–P11 to exhibit deutan-type atypical color vision, whereas P6 and P7 were inferred to exhibit protan-type atypical color vision. Although the Total Error Score (TES) of P4 fell within the range commonly observed among color-neurotypical observers, this participant showed a consistent, axis-specific error pattern. We therefore classified P4 as exhibiting a mild form of atypical color vision with a protan- or deutan-like error pattern. Given the mild nature of this pattern, one possible interpretation is anomalous trichromacy; however, this cannot be confirmed, as the FM100 does not allow a reliable distinction between anomalous trichromacy and dichromacy. The remaining participants (P1, P2, P3, and P5) were inferred to be color-neurotypical, as their error scores fell within the normal range and they did not show axis-specific error patterns characteristic of atypical color vision.

Importantly, these subtype assignments should be regarded as cautious inferences based on characteristic error patterns, rather than definitive physiological diagnoses. In particular, the FM100 does not permit a definitive distinction between protan and deutan subtypes in all cases, and our inferences are therefore intended to be descriptive rather than diagnostic.

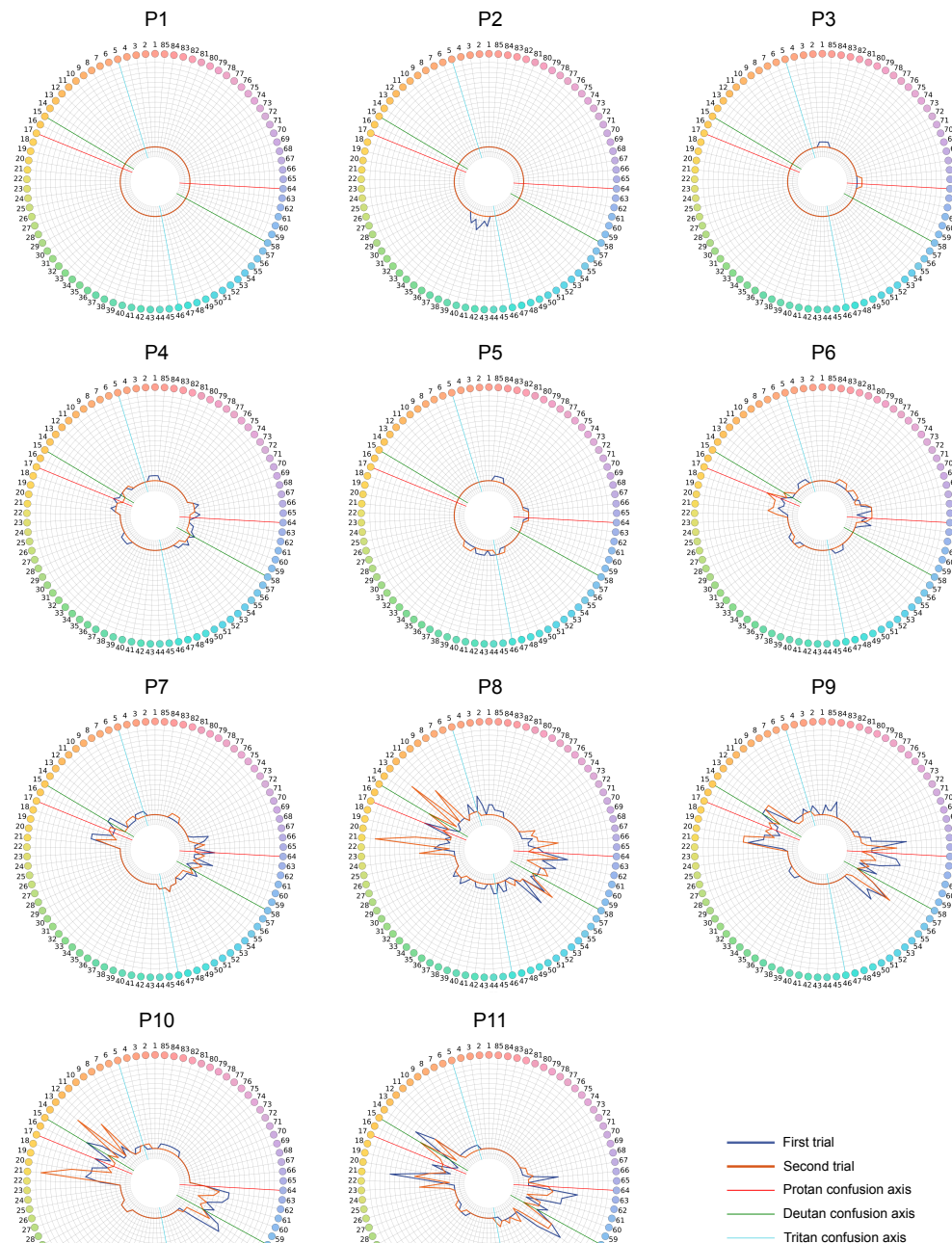


Fig. S1 Error patterns on the FM 100 Hue Test for 11 participants. Each colored circle indicates the color of a cap, and the number attached to each circle denotes the cap number. Blue and orange lines show the per-cap error scores in the first and second trials, respectively (minimum score = 2). Red, green, and skyblue lines indicate the protan, deutan, and tritan confusion axes, respectively.



Fig. S2 Unsupervised alignment of all 11×11 participant pairs. Each matrix shows the optimal transportation plan, in the same format as Fig. 3d.

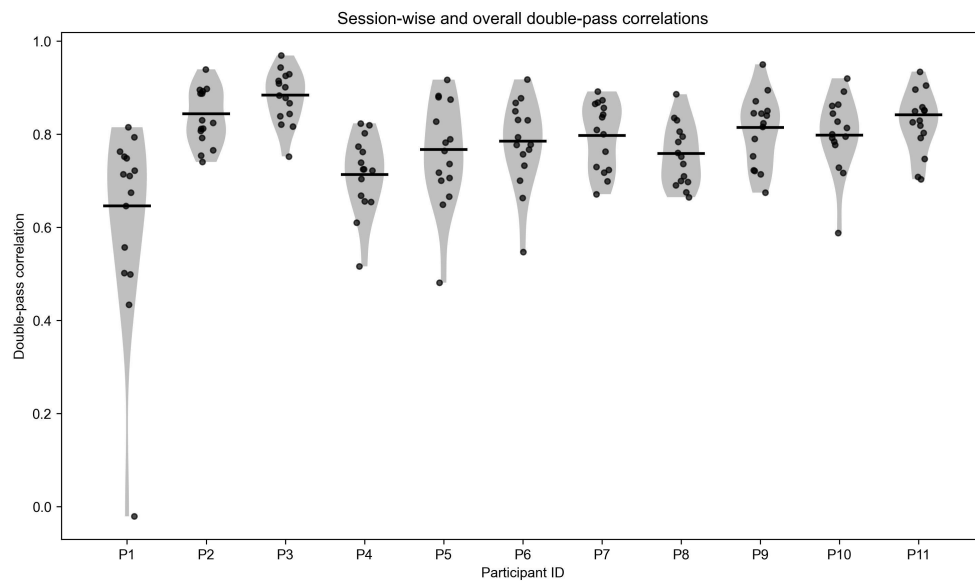


Fig. S3 Double-pass correlations for 11 participants. Each point represents the correlation computed from double-pass pairs within a single session. Horizontal bars indicate the across-session double-pass correlations computed by pooling all double-pass pairs across sessions.